Analytical chemistry 1 for engineers

Homework 5 – Linear regression

姓名： **Name:** **ID number:** **Major:**

H5-1. The sulfate ion concentration in natural water can be determined by measuring the turbidity that results when an excess of $BaCl_2$ is added to a measured quantity of the sample. A turbidimeter, the instrument used for this analysis, is calibrated with a series of standard Na_2SO_4 solutions. The following data were obtained in the calibration. Assume that a linear relationship exists between the instrument reading and the concentration.

a. Calculate the slope and intercept for the best straight line among these points by use of least squares method.

b. Obtain the concentration of sulfate in the unknown sample yielding a turbidimeter reading of 2.84.

SO_4^{2-} (mg/L)	Turbidimeter reading
0.00	0.06
5.00	1.48
10.00	2.28
15.00	3.98
20.00	4.61

a. Calculate the slope and intercept for the best straight line using the data above:

We calculate the sums we need firstly:

$\sum C_i$= 0.00 + 5.00 + 10.00 + 15.00 + 20.00 = 50.00

$\sum R_i$ = 0.06 + 1.48 + 2.28 + 3.98 + 4.61 = 12.41

$\sum C_i R_i$ = 0.00 + 7.40 + 22.80 + 59.70 + 92.20 = 182.10

$\sum (C_i)^2$= 0.00 + 25.0 + 100.0 + 225.0 + 400.0 =750.0

For the slope and intercept for the best straight line, we can use the following formula to calculate *k* and *c*:

$$k = \frac{N\left(\sum x_i y_i\right) - \left(\sum x_i\right)\left(\sum y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2}$$

$$k = \frac{5 \times 182.10 - 50.00 \times 12.41}{5 \times 750.0 - 50.00^2} = 0.232$$

$$c = \frac{\left(\sum x_i^2\right)\left(\sum y_i\right) - \left(\sum x_i\right)\left(\sum x_i y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2}$$

$$c = \frac{750.0 \times 12.41 - 50.00 \times 182.10}{5 \times 750.0 - 50.00^2} = 0.162$$

b. Find the concentration of sulfate yielding a turbidimeter reading of 2.84,

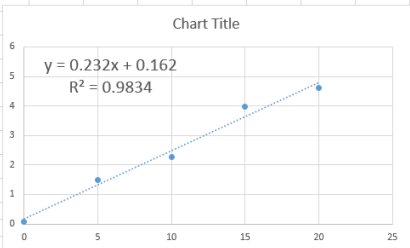
Now we have determined the mathematical relationship between the R and C, we can use it to calculate the sulfate concentration of the sample at a given turbidimeter reading,

$y = kx + c \implies R = 0.232C + 0.162 \quad \implies R = 2.84$

$C = 11.54$ mg/L (keep the same decimal places as the C_i in the table)

*** You also can get the equation from Spreadsheets in Excel**

C	R
0	0.06
5	1.48
10	2.28
15	3.98
20	4.61



H5-2. A 10.00 g sample containing an analyte is transferred to a 250 mL volumetric flask and diluted to the volume using water. When a 10.00 mL aliquot of the resulting solution is diluted to 25.00 mL and it gives a signal of 0.235 (arbitrary unit). A second 10.00 mL aliquot of the solution is spiked with 10.00 mL of a 1.00 ppm standard solution of analyte and diluted to 25.00 mL. the signal for the spiked sample is 0.502. Calculate how much the weight percent of analyte in the original sample?

$[X]_f = [X]_i \frac{V_o}{V} = [X]_i \frac{10.00\ mL}{25.00\ mL}$

$= 0.4000_0\ [Ni^{2+}]_i$

$[S]_f = [S]_i \frac{V_s}{V} = 1.00 \frac{10.00\ mL}{25.00\ mL}$

$= 0.400_0$ ppm

Here, for the signal of 0.235, it comes from the dilution solution of $[X]_f$. So, the formula should be

$$\frac{[X]_f}{[X]_f + [S]_f} = \frac{I_{[X]}}{I_{[X+S)}}$$

$$\frac{0.4000_0 [X]_i}{0.4000_0\ [X]_i + 0.400_0\ \text{ppm}} = \frac{0.235}{0.502}$$

$$[X]_i = 0.880_1\ \text{ppm}$$

So, the mass of analyte in the sample is

0.880_1 ppm × 250 mL = 0.880_1 mg/L × 0.250 L

= 0.220_0 mg

As a result, the weight percent of analyte in the original sample is

$w\% = \frac{0.2200\ \text{mg}}{10.00\ \text{g}} \times 100\% = 2.20 \times 10^{-3}\ \%$

H5-3. To determine the concentration of analyte in a sample, a standard addition was performed. A 5.00 mL portion of sample was analyzed and then successive 0.10 mL spiked of a 600.0 mg/L standard of the analyte were added. The following table shows the results of each spike.

Vs (mL)	0.00	0.10	0.20	0.30
I_{x+s} (Arbitrary unit)	0.119	0.231	0.339	0.442

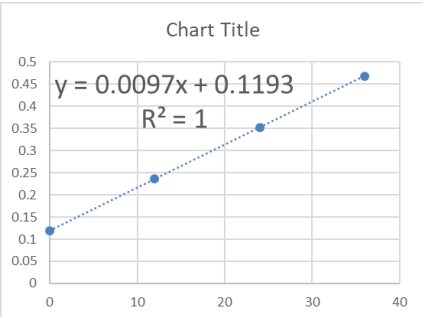
Construct an appropriate standard addition calibration curve and use a linear regression analysis to determine the concentration of analyte in the original sample.

Let's write down all the given data: V_o = 5.00 mL; $[S]_i$ = 600.0 mg/L

First, we need to calculate the values of X-Y. For successive standard addition to a single solution of unknown, the value of Y is $\boldsymbol{I}_{(X+S)}(\frac{V}{V_o})$and X is $[\boldsymbol{S}]_i(\frac{V_s}{V_o})$. The results are listed as below.

Vs (mL)	$[\boldsymbol{S}]_i(\frac{V_s}{V_o})]$ ppm	$I_{(X+S)}$ (arbitrary units)	Signal$\times\frac{V}{V_O}$ (arbitrary units)
0.00	0.00	0.119	$0.119\times\frac{5.00\ mL}{5.00\ mL}$
0.10	$600\times\frac{0.10}{5.00\ mL}$	0.231	$0.231\times\frac{(5.00+0.10)\ mL}{5.00\ mL}$
0.20	$600\times\frac{0.20}{5.00\ mL}$	0.339	$0.339\times\frac{(5.00+0.20)\ mL}{5.00\ mL}$
0.30	$600\times\frac{0.30}{5.00\ mL}$	0.442	$0.442\times\frac{(5.00+0.30)\ mL}{5.00\ mL}$

$[\boldsymbol{S}]_i(\frac{V_s}{V_o})]$/ppm	Signal$\times\frac{V}{V_O}$
0	0.119
12	0.236
24	0.352
36	0.468



$y = 0.0097\,x + 0.1193$ with $R^2 = 1$

Set y = 0, the *x*-intercept is 12.299 mg/L

For a series of standard additions to a single solution of unknown, the X-intercept is the original concentration of analyte in unknown.

So, the analyte in the sample is 12.30 mg/L. (Keep the significant figures as the standard of the analyte of 600.0 mg/L)

H5-4. An experiment was carried out to determine the concentration of sodium ion (Na^+) in a blood sample. Five calibration bottles with 20 mL of the blood sample were added varying amounts of standard NaCl with the concentration of 5.640 M. Then, water was added to obtain in each calibrating bottle with 100 mL solutions. The atomic emission measurement of each solution yielded the signals listed in the table. The question is how much the concentration of the sodium in the blood?

Added standard solution (mL)	**Final volume (mL)**	**Signal (mV)**
0	100	3.13
1	100	5.40
2	100	7.89
3	100	10.30
4	100	12.48

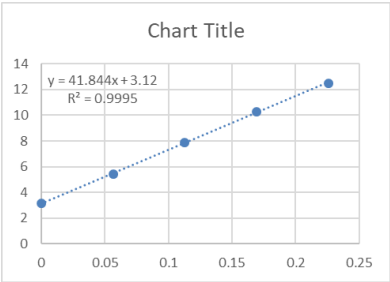
For it is about multiple solution with constant V, we use the formula of

$$I_{(X+S)} = \frac{I_X}{[X]_i}[X]_f + \frac{I_X}{[X]_i}[S]_f$$

The X is $[S]_f$ which means the final concentration of the adding standard (5.640 M of NaCl) in the calibrating flask ($[S]_f = ([S]_i \times Vs/V)$ and the Y is $I_{(X+S)}$.

Y-Signal $\mathbf{I_{(X+S)}}$ mV	Final Volume mL	Added standard solution mL	**X**-$[s]_f$ M
3.13	100	0	0
5.40	100	1	$\frac{1}{100}\times 5.640=0.0564$
7.89	100	2	$\frac{2}{100}\times 5.640=0.1128$
10.30	100	3	$\frac{3}{100}\times 5.640=0.1692$
12.48	100	4	$\frac{4}{100}\times 5.640=0.2256$

Importing the data in the table into Excel, you can get the fitting equation and the squared correlation coefficient (R^2).



The equation determined by least squares method is:

$$Y = 41.844\ X + 3.12$$

so, the X-intercept (when Y=0) is, 3.12/41.844= 0.0745_6 M

As we known, for the multiple solution with constant V, the X-intercept is $[Na^+]_f$, which means the final concentration of sodium ions in the in the calibrating flask (100 mL) is 0.0745_6 M.

According to the dilution factor: $[X]_f = [X]_i \frac{V_o}{V}$

So, the concentration of sodium ions in the blood is

$$[Na^+]_i = \frac{V}{V_o}[Na^+]_f = \frac{100\ mL}{20\ mL} \times 0.07745_6\ M = 0.3728\ M$$

(Keep the significant figures as the standard of NaCl 5.640 M)